

Platform Evolution Roadmap - Sentinel Node

Working product name: **CEL Sentinel Node** - the monitoring backbone of the Celestial mobile solutions line. Strategy: climb the value ladder **environment > power intelligence > connectivity resilience > fleet platform**. Every phase is independently deployable and reuses the phase below it. Baseline: repo v2.0 (CEL-MPT-FW-002, INS-001, QA-001).

Phase map

Phase	Capability added	Key hardware delta	Firmware delta	Sells as
V2.0 (shipped)	Env sentry: temp/RH, leak, door, alert roll-up, Arduino Cloud dashboard	UNO R4 WiFi, BME280, contacts, NEMA 4X node	REV 2.0 field-hardened	Equipment-bay protection
V2.5 - Power Sense	12V bank voltage + current, remote battery-bay temp, 2nd contact (uses spare 22/4 conductor)	Fused sense tap + isolated V front end; hall-effect split-core CT; DS18B20 probes	ADC sampling, SoC-by-voltage table (lead-acid), low-voltage alarm	Battery health + theft/tamper
V3 - Power Intelligence	EcoFlow + charge-source telemetry; energy accounting (Ah in/out)	Victron SmartShunt (VE.Direct) or equiv; BLE to EcoFlow	VE.Direct parser; BLE client; energy integrator	Runtime prediction, generation vs load
V3.5 - Resilient Link	Cellular store-and-forward; GPS breadcrumb	Blues Notecard (LTE-M) + Notecarrier; onboard logging	Offline queue, dual-path publish, position reports	Works anywhere; asset tracking
V4 - Fleet Platform	Multi-unit ops view; REVOLVE + TraceOps + Airtable integration	None per unit (cloud/ops layer)	Fleet IDs, webhook pipe to Command Base	Customer-facing monitored power service

V2.5 flagship design - isolated 12V bank sensing (the promised 'protected front end')

Bank: two 12V lead-acid (810 CCA ea.) in **parallel** = 12V nominal. Design rules:

- **Fuse at the source.** Inline fuse (1-2 A) within inches of the battery tap; sense leads are never unfused.
- **Never divider-direct into the MCU.** Voltage path: fused tap > precision divider > buffered, **galvanically isolated** stage (isolation amplifier, e.g. AMC1311-class, or an isolated I2C ADC module) > MCU ADC/I2C. The trailer bank and the logic node never share a copper path.
- **Current by hall effect, not shunt-in-line with logic.** Split-core hall CT around the main negative lead is non-contact and inherently isolated; clamps on without breaking the circuit. (V3's SmartShunt supersedes this with higher accuracy via its own isolated interface.)
- **SoC by rested voltage table for lead-acid** (12.7 V ~ 100%, 12.4 ~ 75%, 12.2 ~ 50%, 12.0 ~ 25%, alarm <= 11.8 under light load). Note CCA is a cranking rating - runtime math needs Ah/RC from the labels.
- **Alarms:** lowVoltage, highVoltage (charge fault), chargeMissing (no current while generator/PV expected). All roll into systemAlert - dashboard contract unchanged, variables added not changed.

Connectivity tradeoffs

Path	Pros	Cons	Verdict
WiFi (current)	Free; already working; fine at shop/yard	Dies off-network; jobsites rarely have it	Keep as primary at base
BLE (R4's ESP32-S3)	Free; talks to EcoFlow/Victron locally	Short range; not a backhaul	Use as sensor bus (V3), not uplink
Cellular - Blues Notecard LTE-M	Works anywhere; store-and-forward built in; ~\$50 HW + prepaid data incl.; clean Arduino lib	Adds cost per unit; new cloud route (Notehub)	V3.5 backbone for deployed units
LoRa/Meshtastic	No fees; yard-scale mesh	Needs gateway; tiny payloads	Niche - multi-trailer yard telemetry only

Compute platform tradeoffs

Platform	Pros	Cons	Verdict
UNO R4 WiFi (current)	Proven; Qwiic; Arduino Cloud native; crypto onboard	One ADC bank; modest RAM; single unit form	Ride through V3
ESP32-S3 custom carrier	Cheap at qty; BLE+WiFi native; CAN capable	You own the toolchain + cloud auth; more NRE	V4 productization candidate
Arduino Portenta/Opta	Industrial grade; more IO	Cost; overkill for node role	Only if customer spec demands

Fleet + ops integration (V4)

- **Identity:** each node gets a TraceOps asset tag (CEL-AST-xxxx) and a REVOLVE callsign binding (e.g., SC-01 SOL-C / CASSIOPEIA). QR on the enclosure resolves to the unit's live dashboard + doc package.
- **Data pipe:** Arduino Cloud webhooks (or Nohub routes) > Cloudflare Worker > Airtable Command Base (new 14th table: UNIT_TELEMETRY) with alert rows feeding the same lead/dispatch automations as Jotform.
- **Service tiers:** Tier 1 monitor-only (env), Tier 2 power intelligence, Tier 3 monitored power-as-a-service with response SLA - a recurring-revenue line on top of EPC work.
- **Doc plan:** FW-003 (V2.5 firmware), DWG-003 (isolated sense front end), INS-002 (power-sense install), QA-002 (power-sense commissioning), RM revisions here.

Gate to start V2.5: bench prototype passing QA-001 Phases 1-2. Do not stack power sensing on unproven env baseline.